

Imagine-RL: Residual-Confidence-Guided Cross-Attention for World-Model-Augmented VLA Reinforcement Learning

Kejia Hu¹, Wentong Zhai², Bo Zhao³, and Shuai Liang^{4,*}

Abstract—Reliable action evaluation in contact-rich manipulation requires looking beyond the current observation to future visual and contact consequences. Existing noise-space reinforcement learning efficiently steers a frozen Vision-Language-Action (VLA) policy, but its critics largely ignore these consequences. We present Imagine-RL, which augments noise-space VLA post-training with action-conditioned visual-torque imagination. For each candidate action chunk, a frozen visual-torque latent world model (VTLWM) autoregressively predicts compact future representations without pixel reconstruction. A current image-state-action query attends to observed histories and predicted futures, while previous-window prediction residuals provide token-wise confidence priors that suppress unreliable future tokens. By combining current evidence with predicted consequences, the action critic better evaluates candidate actions and supervises the actor, while the VLA and VTLWM remain frozen. Across four real-robot tasks with 50 evaluation trials per task, Imagine-RL uses only 100 RL trajectories and improves the average success rate by (23.6%) over DSRL and by (60%) over VLA baselines.

I. INTRODUCTION

Vision-Language-Action (VLA) models map language, images, and robot states to actions and generalize across tasks and embodiments [1], [2], [3]. However, their predominantly visual observations make contact-rich manipulation difficult. Force-aware VLAs incorporate torque through multimodal fusion or auxiliary prediction [4], [5], but only use it for action generation rather than value estimation.

Reinforcement learning (RL) can adapt VLAs from task-level feedback [6], [7], [8], [9], [10], [11], but real-robot interaction is costly. Moreover, a conventional critic only scores according to current observation and candidate action. This is unreliable when visually similar actions are proposed, although they may lead to either success or failure. In latent-noise VLA-RL, the action critic also supervises the noise critic and actor, making consequence-aware action ranking especially important (Fig. 1).

World models provide a natural source of such look-ahead evidence for reinforcement learning. Existing approaches use visual futures, estimates values, or virtual interactions for policy improvement [12], [13], [14], [15], [16], among which LeWorldModel predicts compact physical representations without heavy pixel reconstruction [17]. Yet force-



Fig. 1. **Predictive critic evaluation.** Both current and predicted future outcomes can be used to distinguish candidate actions, as illustrated by the driving analogy. Without imagining future consequences, two steering actions may appear similarly plausible. By predicting their future trajectories, however, one can identify which action leads to a safer outcome. Similarly, our critic evaluates candidate actions using both the current reward and the predicted future visual-torque outcomes, allowing it to distinguish actions that likely to succeed from those likely to fail.

conditioned imagination for critic learning remains underexplored.

In this paper, we propose **Imagine-RL**, which augments latent-noise VLA-RL with a visual-torque latent world model (VTLWM). For each candidate action, VTLWM predicts action-conditioned future representations. The critic attends to visual and torque histories and futures together with current image-state-action feature. Besides, residuals from the previous prediction window become confidence priors on future-token attention, suppressing unreliable imagination. The frozen VLA and VTLWM require neither backbone fine-tuning nor pixel-level planning, leading to a light solution for contact-rich complicated tasks. Real-robot experiments evaluate on task success rate, world-model prediction, critic discrimination, and shared-candidate selection.

In this paper, we mainly make three contributions:

- We develop Imagine-RL, which integrates action-conditioned visual-torque imagination into lightweight latent-noise VLA post-training.
- We introduce VTLWM, a visual-torque latent world model that predicts action-conditioned future features for value estimation.
- We design a residual-confidence-guided cross-attention [18] critic that down-weights unreliable future predictions while preserving observed history and current evidence.

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II. RELATED WORKS

A. VLA Foundations and Force-Aware VLAs

Vision-language-action models transfer large-scale vision-language and robot-data priors to general-purpose manipulation. Octo and OpenVLA demonstrate broad cross-task transfer [3], [2], while π_0 casts control as flow-based generation of continuous action chunks [1]. Yet these policies remain largely vision-centric, leaving contact phenomena such as resistance, slippage, and jamming only indirectly observable. ForceVLA addresses this gap with a force-aware mixture-of-experts module [5]; TA-VLA instead studies torque adapters and auxiliary torque prediction [4]. These approaches exploit force sensing as a useful policy input.

B. RL Post-Training for VLAs

Strong imitation priors do not eliminate failures caused by distribution shift and compounding execution errors, motivating RL post-training from task-level feedback. $\pi_{0.6}^*$ combines learned values with advantage-conditioned policy learning [6], whereas SimpleVLA-RL studies scalable reinforcement learning for VLAs [7]. Other methods exploit autonomous experience or residual-RL data generation [8], [10], and policy-agnostic RL optimizes actions before distilling them into the policy [11]. DSRL trains a lightweight actor that steers a frozen diffusion/flow policy only through its latent-noise space, instead of tuning the entire VLA [9].

C. World Models for VLA-RL

World models offer a complementary way to reduce physical trial-and-error by anticipating action consequences. World-Env uses a learned environment for VLA post-training [14], while WMPO and GigaBrain-0.5M* perform policy improvement with world-model-generated experience [13], [12]. RISE separates action-conditioned multi-view prediction from progress-value estimation and derives advantages from imagined rollouts [15]. These methods show the value of imagination, but generally rely on visually generated futures. At a lighter level, RL Token demonstrates that compact representations can connect large VLAs to online actor-critic learning [16]. LeWorldModel learns action-conditioned latent dynamics from raw pixels using next-embedding prediction with Gaussian latent regularization, without for pixel reconstruction [17].

III. METHOD

Imagine-RL is a lightweight reinforcement-learning framework that steers a frozen flow-matching VLA through its latent-noise space. A frozen visual–torque world model predicts the consequences of each decoded action chunk, and the resulting predictive values supervise a lightweight noise-space actor.

Fig. 3 presents the Imagine-RL framework. Panel (a) introduces VTLWM; panel (b) shows action-critic learning, noise-critic distillation, and noise-actor optimization; panel (c) constructs predicted features and residual-based prediction confidence; and panel (d) fuses observed and predicted visual–torque features for value estimation.

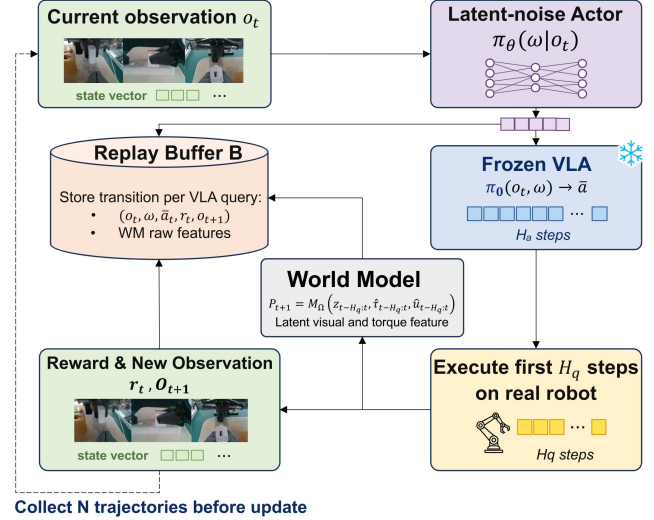


Fig. 2. Replay Buffer Collection of Imagine-RL

Algorithm 1 Imagine-RL with Action-Aligned World-Model Using SAC [19], [20], [21]

Input: frozen VLA policy π_0 , frozen world model M_Ω
offline data $\mathcal{D}_{\text{off}}$
Initialize replay buffer $\mathcal{B} \leftarrow \mathcal{D}_{\text{off}}$, action critic Q_a , target critic $\bar{Q}_a$,
noise critic Q_w , and noise actor π_θ
Initialize feature mapping Φ_ψ with frozen M_Ω
for $t = 1, \dots, T$ **do**
 Sample $(o_t, w_t, \bar{a}_t, r_t, o_{t+1}) \sim \mathcal{B}$
 where $\bar{a}_t = \pi_0(o_t, w_t)$ is the executed action
 1st. Action-critic update
 Sample $w'_{t+1} \sim \pi_\theta(o_{t+1})$ and decode $\bar{a}'_{t+1} \leftarrow \pi_0(o_{t+1}, w'_{t+1})$
 Compute action-aligned critic features
 $h_t^{\bar{a}} \leftarrow \Phi_\psi(o_t, \bar{a}_t)$, $h_{t+1}^{\bar{a}'_{t+1}} \leftarrow \Phi_\psi(o_{t+1}, \bar{a}'_{t+1})$
 Update Q_a by minimizing

$$\mathbb{E} \left[\left(Q_a(o_t, \bar{a}_t; h_t^{\bar{a}}) - r_t - \gamma^{H_q} \bar{Q}_a(o_{t+1}, \bar{a}'_{t+1}; h_{t+1}^{\bar{a}'_{t+1}}) \right)^2 \right]$$

 2nd. Noise-critic update
 Sample $\tilde{w}_t \sim \mathcal{N}(0, I)$ and decode $\tilde{a}_t \leftarrow \pi_0(o_t, \tilde{w}_t)$
 Compute the candidate-aligned critic feature
 $h_t^{\tilde{a}_t} \leftarrow \Phi_\psi(o_t, \tilde{a}_t)$
 Update Q_w by minimizing

$$\mathbb{E} \left[\left(Q_w(o_t, \tilde{w}_t) - Q_a(o_t, \tilde{a}_t; h_t^{\tilde{a}_t}) \right)^2 \right]$$

 3rd. Noise-actor update
 Update π_θ by maximizing

$$\mathbb{E}_{o_t \sim \mathcal{B}} [Q_w(o_t, \pi_\theta(o_t))]$$

end for

A. Latent-Noise VLA-RL

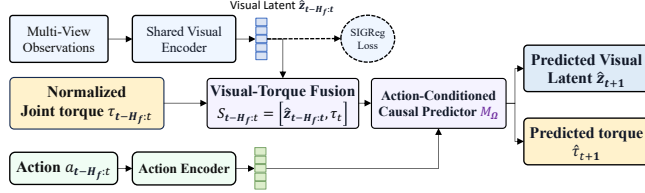
Let o_t denote the current visual observation at decision timestep t . For brevity, robot-state conditioning is implicit in all policy and critic expressions. The frozen VLA π_0 maps the observation and a latent-noise input to a robot action chunk

$$\bar{a}_t = [u_t, u_{t+1}, \dots, u_{t+H_a-1}], \quad \bar{a}_t = \pi_0(o_t, w_t), \quad (1)$$

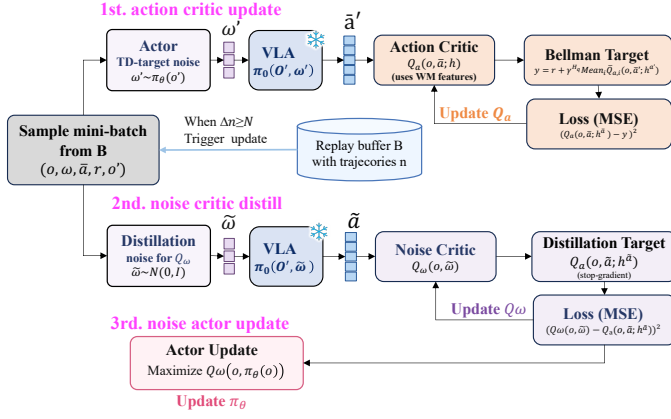
where H_a is the VLA output horizon. Imagine-RL does not learn robot actions directly. Instead, it learns a lightweight latent-noise actor

$$w_t \sim \pi_\theta(w | o_t), \quad (2)$$

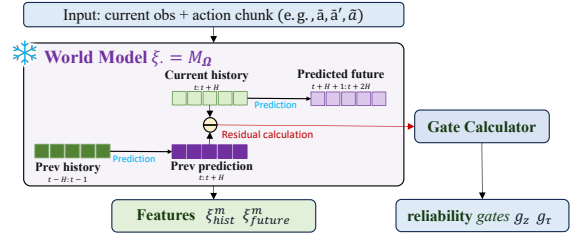
(a) JEPA-style latent world model with torque prediction



(b) Update (sample minibatch from B)



(c) WM feature and gate builder



(d) WM Attention Fusion

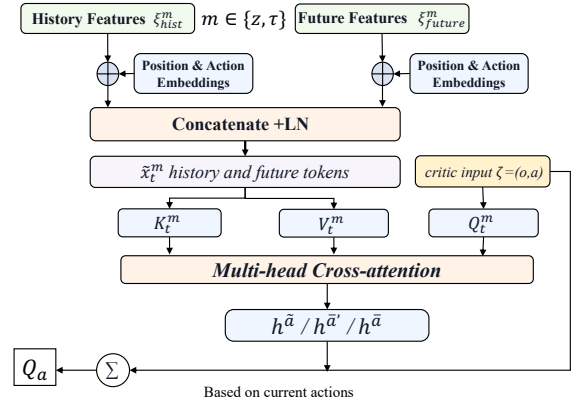


Fig. 3. **Technical realization of the teacher-style predictive critic.** Read panels (a), (c), and (d) as one path: candidate action predicts visual–torque future and is used in confidence-guided attention to give teacher action value. Panel (b) then shows how this value teaches the noise critic and actor. Thus, the action critic scores what the robot is doing now together with what that action is predicted to cause.

which steers the frozen VLA through its noise input.

We use two critics as shown in Fig. 3(b). The action critic $Q_a(o, \tilde{a}; h^{\tilde{a}})$ evaluates decoded robot-action chunks and according to historical-current observations and world-model-generated future visual-torque feature $h^{\tilde{a}}$, given $\tilde{a}_t$. Then latent-noise critic $Q_w(o, w)$ is distilled from $Q_a(o, \pi_\theta(o, w))$. This keeps the deployed policy lightweight: during RL, only noise actor π_θ is tuning and the base VLA is frozen.

The replay buffer, as shown in Fig. 2, save one transition per VLA query, with chunk-level reward r_t and discount γ^{H_q} . For $w_t \sim \pi_\theta(\cdot | o_t)$ and $\tilde{a}_t = \pi_\theta(o_t, w_t)$, we train multiple Q networks and get the mean in order to avoid bias

$$y_t = r_t + \gamma^{H_q} \frac{1}{n} \sum_{i=1}^n \bar{Q}_{a,i} \left(o_{t+1}, \tilde{a}_{t+1}; h_{t+1}^{\tilde{a}'} \right), \quad (3)$$

where n denotes the number of Q. The exact contents of $h_t^{\tilde{a}}$ are defined in Eq.(12).

The noise critic is trained by querying the frozen VLA with sampled latent noise $\tilde{w}$:

$$\min_{Q_w} \mathbb{E} \left[\left(Q_w(o_t, \tilde{w}_t) - Q_a(o_t, \tilde{a}_t; h_{t+1}^{\tilde{a}'}) \right)^2 \right], \quad (4)$$

$$\tilde{a} = \pi_\theta(o_t, \tilde{w}).$$

The latent-noise actor is then updated to maximize $Q_w(o, \pi_\theta(o))$, with the usual entropy regularization in imple-

mentation. The complete training procedure is summarized in Algorithm 1.

B. Force-Aware Latent World Model

To achieve better performance on contact-rich manipulation, Imagine-RL conditions the action critic on predictive features from a visual–torque latent world model (VTLWM). Extending LeWorldModel [17], VTLWM jointly predicts visual and torque representations without pixel reconstruction: a shared encoder fuses multi-view images into z_t , while torque and action inputs are normalized as $\hat{\tau}_t$ and a_t . We define

$$x_t = [z_t, \hat{\tau}_t], \quad e_t^a = E_a(a_t). \quad (5)$$

We use a fixed one-step model with history length $H = 15$. Given $H + 1$ consecutive samples, the causal predictor uses the first H state–action pairs to predict $H + 1$ state:

$$P_H = M_\Omega(X_{0:H-1}, E_{a_{0:H-1}}), \text{ where}$$

$$X_{0:H-1} = [x_0, \dots, x_{H-1}], E_{a_{0:H-1}} = [E_{a_0}, \dots, E_{a_{H-1}}] \quad (6)$$

Here, M_Ω is our world model, VTLWM. During rollout, the final prediction P_H is appended to the context and the inference window is shifted by one step, yielding an

autoregressive future sequence. VTLWM training loss is:

$$\mathcal{L}_{\text{WM}} = \text{MSE}(P_{1:H}^z, Z_{1:H}) + \lambda_\tau \text{SmoothL1}(P_{1:H}^\tau, T_{1:H}) + \lambda_{\text{sig}} \mathcal{L}_{\text{SIGReg}} \quad (7)$$

where $Z_{1:H}$ and $T_{1:H}$ are the encoded visual and normalized force ground-truth features.

During RL training, VTLWM is frozen and only generates critic features. For each modality $m \in \{z, \tau\}$, the feature builder returns: (1) the current history $X_t^{m, \text{hist}}$, (2) a future prediction $X_t^{m, \text{fut}}(a_t)$ conditioned on the evaluated action, (3) a previous-window prediction $P_t^{m, \text{prev}}$.

$$\begin{aligned} R_{t,j}^m &= X_{t,j}^{m, \text{hist}} - P_{t,j}^{m, \text{prev}}, \\ \mathcal{S}_t^m(a_t) &= \{X_t^{m, \text{hist}}, X_t^{m, \text{fut}}(a_t), \\ &\quad P_t^{m, \text{prev}}, R_t^m\}. \end{aligned} \quad (8)$$

History state, future prediction, and history-state-prediction residual masks indicate which positions are temporally aligned. $j \in [0 : H_f - 1]$ is the index within each prediction horizon. We use $H = 15$ history positions and recursively apply the one-step predictor for $H_f = 10$ future positions. The remaining future positions are padded and excluded by the future mask.

C. Confidence-Guided Cross-Attention Critic

In order to reasonably utilize information, we design a Residual-Confidence-Guided Cross-Attention [22], [23] Critic, that scores a candidate action by selecting relevant visual and torque information from observed history and predicted futures. Attention determines what to retrieve, according to the prediction-error confidence that indicating how much to trust the future. The retrieved information augments extra feature besides the current observation and action for value estimation. Here we describe one modal m as example.

a) Query and tokens: Let $\zeta_t = (o_t, a_t)$ denote the current visual observation paired with the candidate action chunk ($\bar{a}_t$ or $\tilde{a}_t$ above). A learned query map f_q converts ζ_t as a $d = 256$ -dimensional vector q_t :

$$q_t = f_q(\zeta_t). \quad (9)$$

The world-model future is generated under that same action.

In Fig. 3(d), ξ_{hist}^m and ξ_{future}^m denote projected $X_t^{m, \text{hist}}$ and $X_t^{m, \text{fut}}(a_t)$; adding position and action embeddings produces the corresponding attention tokens. $P_t^{m, \text{prev}}$ is position-aligned with $X_t^{m, \text{hist}}$ and is used only for residual confidence estimation. Here $D_z = 192$ and $D_\tau = 14$; stacking the H history and H future tokens gives $\tilde{X}_t^m$, with only H_f valid future positions.

b) Prediction reliability: The previous-window features from the current timestep t , the historically predicted feature $P_{t,j}^{m, \text{prev}}$ and the observed history $X_{t,j}^{m, \text{hist}}$, indexed by j , are at the same timestamp; their difference is the residual $R_{t,j}^m$ in Eq.(8), representing the WM generation reliability that persists across nearby windows. With the mean squared

feature magnitude $E_m(x) = \|x\|_2^2/D_m$, the relative error $\rho_{t,j}^m$ and generation reliability confidence $\alpha_{t,j}^m$ are

$$\begin{aligned} \rho_{t,j}^m &= \frac{E_m(R_{t,j}^m)}{\frac{1}{2}[E_m(X_{t,j}^{m, \text{hist}}) + E_m(P_{t,j}^{m, \text{prev}})] + \epsilon}, \\ \alpha_{t,j}^m &= \max(\alpha_{\min}, e^{-\beta \rho_{t,j}^m}). \end{aligned} \quad (10)$$

This means smaller relative errors yield higher confidence. Here $\epsilon = 10^{-6}$ prevents division by zero, $\beta = 1$ controls sensitivity, and $\alpha_{\min} = 0.05$ is the confidence floor. Unavailable residuals receive $\alpha_{\min}$ and $\rho = 4$ for the statistics below.

c) From token relevance to context: Separate learned projections produce a modality-specific query $\mathbf{Q}_t^m = q_t W_Q^m$, matching keys $\mathbf{K}_t^m = \tilde{X}_t^m W_K^m$, and content values $\mathbf{V}_t^m = \tilde{X}_t^m W_V^m$. Keys and values represent the same tokens: keys determine relevance, while values supply the information retrieved, Fig. 3(d). Q_a denotes action value. Each modality has its own projection parameters; affine biases are omitted.

The projected width d is split into four heads indexed by ℓ , each with dimension $d_\ell = 64$. The bias vector $\mathbf{B}_t^m$ has one entry per token: 0 for valid history, $\log \alpha_{t,j}^m$ for valid future step j , and $-\infty$ for invalid slots. Each head computes

$$\begin{aligned} \mathbf{A}_{t,\ell}^m &= \text{softmax}\left(\frac{\mathbf{Q}_{t,\ell}^m (\mathbf{K}_{t,\ell}^m)^\top}{\sqrt{d_h}} + \mathbf{B}_t^m\right), \\ c_{t,\ell}^m &= \mathbf{A}_{t,\ell}^m \mathbf{V}_{t,\ell}^m, \\ c_t^m &= \text{LN}(\text{Concat}_{\ell=1}^4(c_{t,\ell}^m) W_O^m). \end{aligned} \quad (11)$$

Query-key dot products measure relevance; softmax converts the biased scores into token weights $\mathbf{A}_{t,\ell}^m$ summing to one. Their weighted sum of token values is the single-head context $c_{t,\ell}^m \in \mathbb{R}^{64}$. Concatenating the four contexts and applying the learned output projection W_O^m and layer normalization (LN) yields $c_t^m \in \mathbb{R}^{256}$. Adding $\log \alpha$ discounts a future token's unnormalized weight by α , combining relevance with reliability. Entirely invalid sequences produce zero context.

d) From contexts to action value: Contexts c_t^z, c_t^τ summarize retrieved information. The four reliability statistics g_t^m are the mean, minimum, and maximum confidence and mean relative error over valid future slots, or zeros if none are valid. The core feature fusion is

$$\begin{aligned} h_t^a &= [c_t^z, c_t^\tau, g_t^z, g_t^\tau], \\ Q_a(o_t, a_t; h_t^a) &= \mathcal{Q}([\zeta_t, h_t^a]). \end{aligned} \quad (12)$$

$\mathcal{Q}$ is a MLP critic outputting the Q value for policy optimization; brackets concatenate the corresponding feature representations. Thus, $\zeta_t = (o_t, a_t)$ supplies the current observation and action, while h_t^a adds the retrieved world-model contexts and their reliability. Algorithm 1 denotes this feature construction by Φ_ψ , with learned critic-side parameters ψ and implicit history inputs. We design multiple critics for stability and each learns its own updating; VTLWM remains frozen.

IV. EXPERIMENTS

We conduct real-robot experiments to answer three questions: (1) whether Imagine-RL improves task success

rate over pretrained and RL-post-trained VLA baselines; (2) whether VTLWM predicts contact-relevant visual and torque dynamics; and (3) whether residual-confidence-guided visual–torque attention improves critic discrimination.

A. Experimental Setup / Results

a) *Real-Robot Tasks.*: We evaluate all methods on four contact-rich manipulation tasks, illustrated in Fig. 4(a). The tasks include Socket Plug-in, Press-button, USB Plug-in, and Peg-in-hole, Fig. 5. Each task is dual arm collaborative manipulation task, including multiple stage within one task, including pressing, picking up, pressing, pushing, etc. For each task, we randomize object position and orientation within the same range during evaluation.

b) *Baselines.*: We compare the following four methods:

- π_0 : the pretrained visual VLA without task-specific reinforcement learning;
- **TA-VLA**: a torque-aware VLA that incorporates force physical feedback/prediction;
- **DSRL**: latent-noise reinforcement learning with a world-model-free critic;
- **Imagine-RL**: our method, which augments latent-noise RL with residual-confidence-guided cross-attention over visual and torque histories and candidate-action-conditioned futures through our Latent World Model.

c) *Training Protocol.*: For each task, the base π_0 is pretrained with 100 trajectories. Imagine-RL and DSRL use the same pretrained VLA backbone and an initial warm-up buffer of 50 trajectories. Both methods receive the same amount of online interaction. At each non-terminal query transition, the reward is -1 ; successful termination gives $+5$, ordinary failed truncation gives -1 , and a severe absorbing failure gives -50 . We use two NVIDIA RTX 3090 GPUs to host the remote base VLA (π_0) server, while the RL components are trained locally on a single NVIDIA RTX 3090 GPU.

d) *Implementation Details.*: Our robotic platform is a PIPER robotic arm equipped with Intel RealSense D435i cameras. The frozen VLA outputs an action chunk of length $H_a = 30$, of which the first $H_q = 10$ actions are executed before replanning. VTLWM uses a history length $H = 15$ and a one-step predictor, recursively rolled out for $H_f = 10$ valid future visual and torque states at 30 Hz. The visual and torque token dimensions are 192 and 14, respectively. Cross-attention has width 256 and four heads, with no attention dropout. The action critic uses an ensemble of ten Q-functions and mean target reduction. We use a batch size of 64, a per-control-step discount of 0.999, actor learning rate 5×10^{-5} , critic-encoder learning rate 10^{-4} , and critic-head/fusion learning rate 3×10^{-4} . RGB inputs from the left and right wrist cameras and the top-view camera are resized to 128×128 .

Before the first RL update, we collect 50 pure- π_0 trajectories and run at most 5000 initial gradient steps. Thereafter, RL training is performed after every ten newly collected trajectories. All methods use identical RL hyperparameters unless otherwise stated.

e) *Results.*: We evaluate each method for 50 trials per task, with the training procedure of every 10 episodes updating once. The initial object poses and task conditions are kept the same for all methods. Success rates are shown in Fig. 6. For the two imitation methods, our method achieves an average relative success rate rise of 60% compared to π_0 and 60% compared to TA-VLA. For the reinforcement learning baseline DSRL, we achieve a rise of 23.6%, within only 100 episodes updating (50 episodes warm-up, $10 * 5$ episodes updates).

To examine performance with additional interaction, we continue training Imagine-RL on the *press-button* task until the online buffer up to 100 trajectories ($10 * 10$ episodes updating) after warm-up training. Fig. 7 shows the resulting success rate as the amount of online data increases, which achieves $+46\%$ success rate increment for task **Press-button**, within only $10 * 10$ episodes RL updating after 50 episodes warm up training, showing the capability of our method can improve fast within only small numbers RL trials.

B. World-Model Prediction Analysis

We next test whether VTLWM captures the key future dynamics required by the critic. We evaluate the frozen world model on held-out-robot sequences.

a) *Visual Future Prediction.*: Fig. 4(b) shows the qualitative visual feature prediction performance of VTLWM on four tasks (Socket Plug-in, Press-button, USB Plug-in, Peg-in-hole), comparing predicted pixel visual futures decoded by a trained decoder with ground-truth observations. The decoder is not used by the critic or the deployed policy. VTLWM autoregressively predicted future visual representation $O_{t+H_f}^P$ and their corresponding ground-truth observations $O_{t+H_f}^{GT}$, conditioned with initial visual observation O_t^{GT} , torque T_t^{GT} and action $A_{t:t+(H_f-1)}$, here $H_f = 10$. As shown in Fig. 4(b), VTLWM captures gripper-object interaction showing consistency with ground-truth as the action proceeding within prediction horizon. Although prediction quality gradually decreases with rollout length resulting from the compounding error through VTLWM autoregressive inference, the predicted representations still preserve the task-relevant temporal and spatial features between the robot and the environment as action evolves, which is determinative information for critic estimation.

b) *Torque Future Prediction.*: Fig. 8 shows the qualitative torque feature prediction performance of VTLWM on four tasks (Socket Plug-in, Press-button, USB Plug-in, Peg-in-hole), compared with ground-truth torque values on 12 ($6*2$) joints and 2 grippers.

TABLE I
COSINE SIMILARITY OF VISUAL PREDICTIONS

Method	Socket	Press	USB	Peg	Avg.
Visual	0.955	0.915	0.881	0.901	0.913

VTLWM autoregressively predicted future torque $T_{t+H_f}^P$ and their corresponding ground-truth torque $T_{t+H_f}^{GT}$, con-

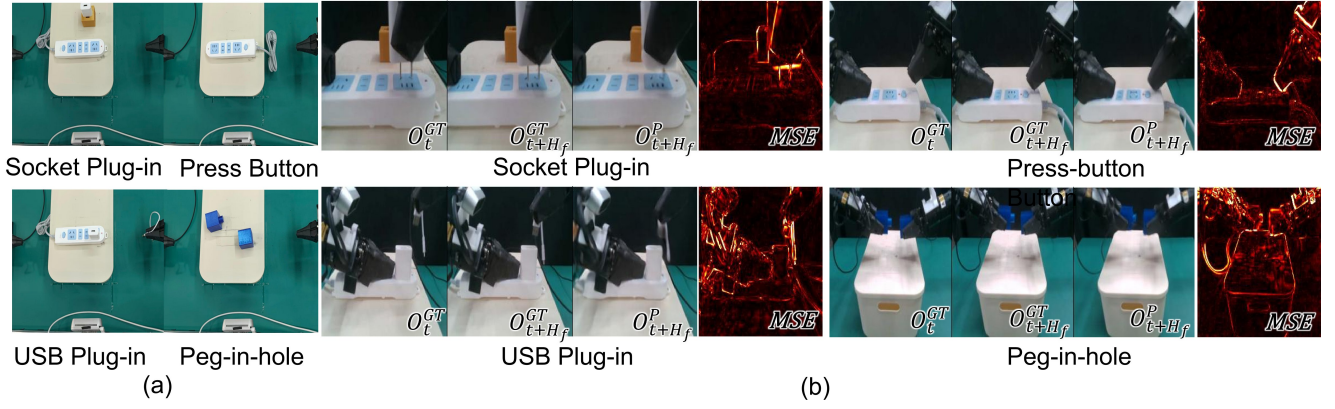


Fig. 4. **Real-robot settings and qualitative results of action-conditioned visual prediction from our Latent World Model (key frames).** (a) Four contact-rich settings (Socket Plug-in, Press-button, USB Plug-in, Peg-in-hole). (b) VTLWM autoregressively predicted future representations $O_{t+H_f}^P$ and their corresponding ground-truth observations $O_{t+H_f}^{GT}$, conditioned with initial observation O_t^{GT} and action $A_{t:t+(H_f-1)}$, here $H_f = 10$. The MSE between $O_{t+H_f}^{GT}$ and $O_{t+H_f}^P$ are calculated and shown at end of each task.

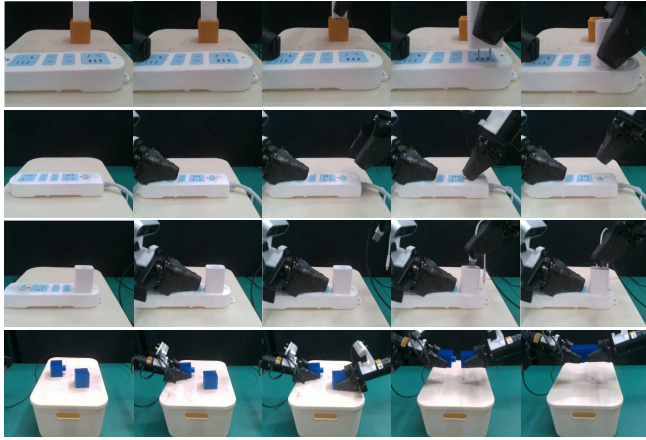


Fig. 5. **Qualitative experiment results of four real robot tasks (Socket Plug-in, Press-button, USB Plug-in, Peg-in-hole).** Each task is a dual-arm collaborative manipulation task, including multiple stages within a single task, such as pressing, picking up, pressing again, and pushing.

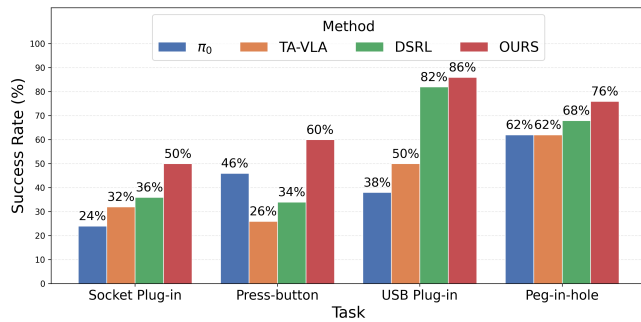


Fig. 6. **Success rates on the four real-robot tasks (Socket Plug-in, Press-button, USB Plug-in, Peg-in-hole, each task 50 trials), 50 episodes RL training (every 10 episodes updating once), after 50 episodes warm-up RL training.**

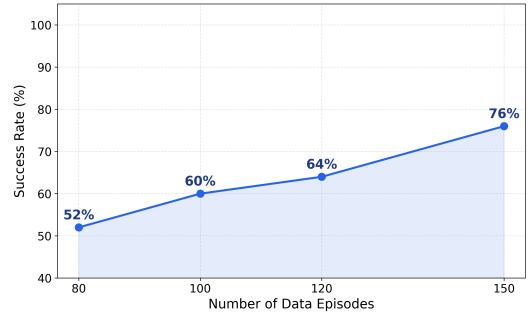


Fig. 7. **Success rate with increasing online interaction data.**

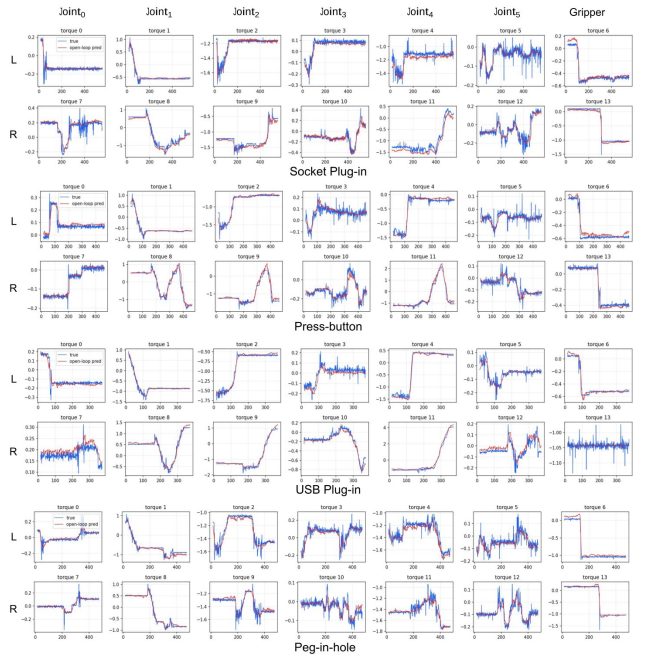


Fig. 8. **Torque prediction on held-out real-robot trajectories on four tasks (Socket Plug-in, Press-button, USB Plug-in, Peg-in-hole).** Red lines denote ground-truth measured torque and blue lines denote VTLWM predictions. L and R represents the left and right robot arms.

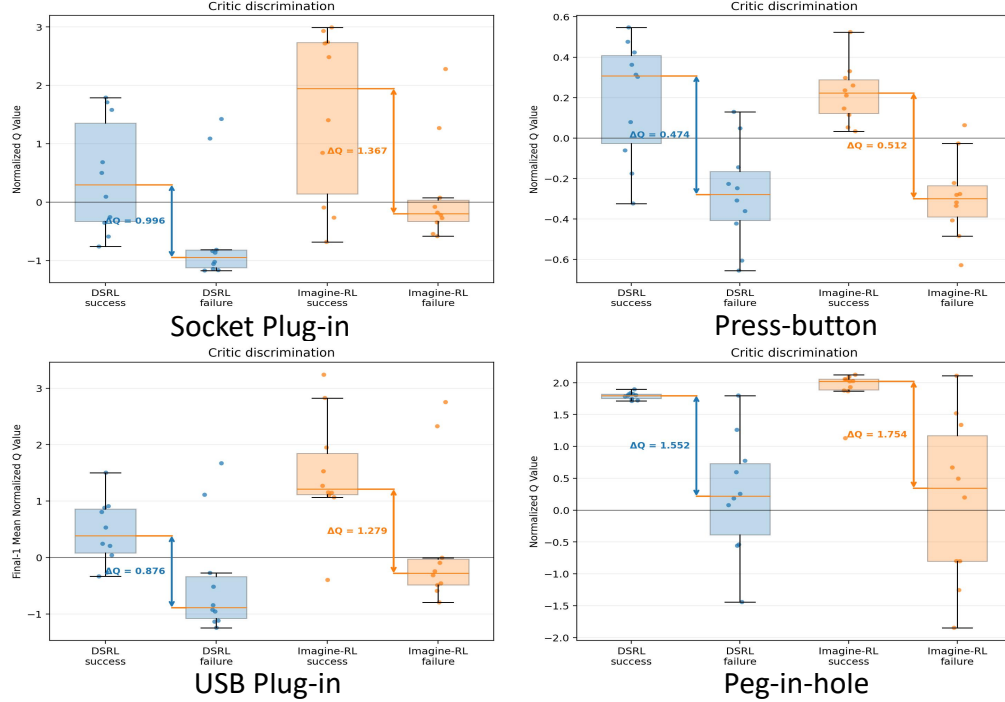


Fig. 9. **Critic discrimination analysis.** Results on Socket Plug-in, Press-button, USB Plug-in, and Peg-in-hole. For each task, DSRL and Imagine-RL score the same executed action chunks from 10 held-out successful and 10 failed trajectories under the same reward setting. Q-values are z-score standardized separately for each critic within each task, and boxes summarize the trajectory-level distributions. For all four tasks, each dot is the standardized Q-value at the final retained VLA query step of one trajectory. The annotated margin is $\Delta Q = \mathbb{E}[Q | \text{success}] - \mathbb{E}[Q | \text{failure}]$. The DSRL and Imagine-RL margins are, respectively, 0.996 and 1.367 on Socket Plug-in, 0.474 and 0.512 on Press-button, 0.876 and 1.279 on USB Plug-in, and 1.552 and 1.754 on Peg-in-hole. These correspond to relative increases of 37.2%, 8.0%, 46.0%, and 13.0%, respectively, demonstrating consistently stronger critic discrimination by Imagine-RL.

TABLE II
TORQUE PREDICTION PERFORMANCE ON MAE, MSE, RMSE OF FOUR
REAL ROBOT TASK

Method	Socket	Press	USB	Peg	Avg.
MAE	0.0483	0.0455	0.0449	0.0378	0.0441
MSE	0.0059	0.0070	0.0053	0.0035	0.0054
RMSE	0.0770	0.0834	0.0731	0.0589	0.073

ditioned with initial visual observation O_t^{GT} , torque T_t^{GT} and action $A_{t:t+(H_f-1)}$, here $H_f = 10$. The predicted force signals are able to effectively capture the onset and temporal evolution of contact events and key robot motion, providing extra action trigger signal for critic evaluation, and complementing information that is only weakly observable from RGB inputs. For example, during the press-button task, whether a press is effective can be determined by sensing the vertical elastic force, which continuously increases with the pressing depth and suddenly disappears when the button locks. Such contact dynamics are difficult to capture using visual sensing alone.

We further evaluate VTLWM quantitatively. All the data are from the 50 warm-up and 10*5 RL trials from the frozen pre-trained VTLWM. We calculate the mean visual cosine similarities of the latent embedding between the prediction and ground-truth across the whole four tasks, as reported in

Table I. From Table I we can find that the visual cosine similarities between ground truth and VTLWM predicted embedding are very close to 1, illustrating that our VTLWM is able to capture the dynamics and predict the reasonable future evolution in advance, providing key information for critic. Furthermore, we calculate the MAE, MSE, RMSE of torque prediction on each task, as shown in Table II. VTLWM achieves relatively high predictive accuracy, with an MAE of 0.0441, corresponding to only 1.00% of the maximum value (± 4.3799).

C. Critic Discrimination

We examine whether the action critic can distinguish executed actions from successful and failed interactions under the same reward definition and scale. For each of the four tasks, we use the same held-out set of 10 successful and 10 failed trajectories for DSRL and Imagine-RL. At every retained VLA query step, both critics receive the same observation and score the same executed action chunk of length H_q ; Imagine-RL additionally conditions its evaluation on the action-conditioned visual-torque features predicted by VTLWM. Thus, the comparison isolates how well the learned value functions discriminate action outcomes rather than differences in the evaluated actions or rewards.

Because the two critics are trained independently and their raw Q-values need not share a common scale, within each

task we standardize each critic as $\bar{Q} = (Q - \mu_Q)/(\sigma_Q + \epsilon)$, with μ_Q and σ_Q computed over all query-step Q-values from both outcome sets. Each point in Fig. 9 is the standardized Q-value at the final retained VLA query step of one trajectory. We use this same terminal-interaction statistic for all four tasks and quantify discrimination as $\Delta Q = \mathbb{E}[\bar{Q} \mid \text{success}] - \mathbb{E}[\bar{Q} \mid \text{failure}]$, where $\bar{Q}$ denotes the within-critic standardized value. A larger ΔQ therefore indicates a stronger separation between actions associated with successful and failed outcomes. Imagine-RL consistently produces a larger margin than DSRL on all four tasks. Averaged across the four tasks, the discrimination margin increases from **0.975** for DSRL to **1.228** for Imagine-RL, corresponding to a **26.0%** relative improvement. These results show that the VTLWM-conditioned critic assigns more distinctive values to successful and failed interaction actions under an identical reward setting.

V. CONCLUSION

We presented Imagine-RL, a lightweight approach for adapting a frozen VLA to contact-rich manipulation through latent-noise reinforcement learning. A visual–torque latent world model (VTLWM) predicts the consequences of each action chunk evaluated by the critic. The action critic uses the current image–state–action representation to query visual and torque history/future tokens, while verified past prediction residuals suppress unreliable future tokens through an additive attention prior. This action-aligned predictive critic provides the teacher values used to train the lightweight noise critic and actor without fine-tuning the VLA backbone or performing pixel-level planning. Our real-robot evaluations examine the resulting task performance, visual–torque latent world model prediction performance and critic discrimination. Experiment results show that our Imagine-RL method achieves faster convergence speed and higher success rate compared with baselines. Future work will investigate continual world-model adaptation under policy-induced distribution shift, multi-timescale prediction of delayed contact outcomes, and uncertainty-aware exploration with safe fallback to the frozen VLA prior.

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